

Web Engineering Lab Report Portfolio

Week 01 to Week 03 Lab Submissions

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CMS ID: 023-24-0118	Lab: Weeks 01 to 03 Combined

Portfolio Overview

This document contains my completed lab submissions for Web Engineering (Weeks 1 to 3). It includes all in-class activities, source code, DevTools testing, and homework tasks for:

- **Week 01:** How the Web Works, Toolchain Setup & DevTools Drills
- **Week 02:** Semantic HTML5, Forms, Data Tables & Validation
- **Week 03:** CSS Fundamentals, Box Model, Selectors & Specificity

Web Engineering — Lab 01 Report

How the Web Works, Tools Setup & DevTools Basics

Student Name: Muhammad Tayyab	Course: Web Engineering (5th Semester)
CMS ID: 023-24-0118	Lab: Week 01: How the Web Works

1. Lab Summary & What We Covered

In this first lab, we learned the basics of how the web works, how browsers communicate with servers, how URLs are structured, and how to use Chrome DevTools. We also installed and set up Node.js, npm, Git, and VS Code extensions (Live Server and Prettier), and created our first HTML page.

2. Warm-up & Client vs Server (Activity A)

Warm-up: I opened YouTube/Google in Chrome, pressed F12 to open DevTools, and checked the Network tab. I saw around 40 requests loading, with status 200 OK for HTML and scripts.

Client vs Server: The web works as a conversation between two sides:

- **Client (Frontend):** Runs in the browser (HTML, CSS, JavaScript). It displays the page to the user and sends requests.
- **Server (Backend):** Runs on another computer (Node.js, Express, Database). It handles requests, logic, and sends back data.

3. Anatomy of a URL (Activity B)

We broke down the example URL `https://api.example.com:443/products/42?sort=price#reviews`:

Part	Value	What it means
Protocol	https://	Secure HTTP protocol with TLS encryption.
Host / Domain	api.example.com	The server address (resolved to IP via DNS).
Port	:443	Port number (443 for HTTPS, 80 for HTTP, 5500 for Live Server).
Path	/products/42	Path to the specific resource on the server.
Query	?sort=price	Extra parameters sent to server for filtering or search.
Fragment	#reviews	Jump link to an anchor on the page (client-side only).

4. HTTP Status Codes (Activity C)

Family	Codes	Meaning	What to do
2xx	200 OK, 201	Request succeeded.	Page/data loaded normally.
3xx	301, 304	Redirect / Loaded from cache.	Browser uses cached file or follows new link.
4xx	400, 404	Client error (wrong URL / bad request).	Fix the URL or client request.
5xx	500, 502	Server error (backend crashed).	Fix the server code / database.

5. Environment Setup Output (Activity D & Homework 1)

Terminal output
<pre>\$ node -v v20.18.0 \$ npm -v 10.8.2 \$ git --version git version 2.43.0 \$ git config --global user.name "Muhammad Tayyab" \$ git config --global user.email "tayyab@example.com"</pre>

6. In-Class Lab: First Page (`hello-web/index.html`)

hello-web/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Hello Web - Muhammad Tayyab</title>
</head>
<body>
  <h1>Hello, Web Engineering</h1>
  <p>My name is <strong>Muhammad Tayyab</strong> (CMS ID: 023-24-0118) and this is my first page.</p>
  <p>I am learning web development to build full-stack web applications.</p>

  <h2>Three things I want to build this semester:</h2>
  <ul>

    <li>A responsive personal portfolio website</li>
    <li>A full-stack CRUD web app</li>
    <li>A clean frontend interface with React</li>
  </ul>

  <button id="greet">Click me</button>

  <script>
    document.getElementById("greet").addEventListener("click", () => {
      alert("The button works! - Muhammad Tayyab");
    });
  </script>
</body>
</html>
```

7. DevTools Drills & Error Explanation (Activity F)

- **Network Tab:** Reloaded the Live Server page (<http://127.0.0.1:5500/hello-web/index.html>) and verified that index.html loaded with status **200 OK**.
- **Console Tab:** Typed document.title in console and it returned 'Hello Web - Muhammad Tayyab'.
- **Elements Tab:** Double clicked the <h1> tag in Elements and changed the text. When I reloaded the page, the edit was gone because DevTools only changes the temporary DOM in browser memory, not the actual file on my laptop.
- **Deliberate Error Test:** Typed console.log(testVariable) in console.

Error: Uncaught ReferenceError: testVariable is not defined

Explanation: **The error happened because the variable 'testVariable' was never created or declared before trying to use it.**

8. Homework Task 2: Network Detective

I opened github.com and inspected the Network tab:

1. **Total requests:** 42 requests loaded on the page.
2. **Largest file:** A JavaScript bundle file main.js (around 380 KB).
3. **Non-200 request:** Found GET /favicon.ico returning status **304 Not Modified**. This happened because the browser already had the icon cached from a previous visit, so the server told it to use the cached copy instead of downloading it again.

Web Engineering — Lab 02 Report

HTML Deep Dive — Semantic Elements, Forms & Tables

Student Name: Muhammad Tayyab	Course: Web Engineering (5th Semester)
CMS ID: 023-24-0118	Lab: Week 02: HTML Deep Dive

1. Lab Summary

In Week 02, we learned how to use semantic HTML5 tags instead of just using `<div>` tags everywhere. We practiced building an accessible profile page with internal jump links, adding images with `<figure>` and `<figcaption>`, creating forms with proper `<label>` and `<input>` types, building data tables, and passing the W3C validator.

2. Semantic HTML Elements Used

- `<header>`: Top banner with my name, tagline, and navigation.
- `<nav>`: Contains navigation links (`#about`, `#skills`, `#contact`).
- `<main>`: The main content of the page (used only once per page).
- `<article>`: Self-contained section for the About Me card.
- `<section>`: Thematic sections for Skills and Contact form.
- `<aside>`: Sidebar / extra info about lab timings.
- `<footer>`: Footer with copyright and social links.

3. In-Class Lab Deliverable (Week 02/profile.html)

Week 02/profile.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Profile page of Muhammad Tayyab" />
  <title>Muhammad Tayyab - Profile</title>
</head>
<body>
  <!-- Header -->
  <header>
    <h1>Muhammad Tayyab</h1>
    <p>BS Computer Science / Software Engineering Student (CMS ID: 023-24-0118)</p>
    <nav>

      <ul>
        <li><a href="#about">About</a></li>
        <li><a href="#skills">Skills</a></li>
        <li><a href="#contact">Contact</a></li>
        <li><a href="contact.html">Contact Page</a></li>
      </ul>
    </nav>
  </header>

  <!-- Main Content -->
  <main>
    <!-- About Section -->
    <article id="about" class="card">
      <h2>About Me</h2>

      <p>Hi, I am Muhammad Tayyab. I am learning web engineering this semester.</p>
      <p class="muted">Currently learning HTML5, CSS3, and JavaScript basics.</p>
      <figure>
        
        <figcaption>My coding workspace setup.</figcaption>
      </figure>
    </article>

    <!-- Skills Table Section -->
    <section id="skills" class="card">
      <h2>My Tools & Skills</h2>
      <table>
        <caption>Tools and Comfort Level</caption>
        <thead>

          <tr>
            <th scope="col">Tool</th>
            <th scope="col">Comfort</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <th scope="row">HTML5</th>
            <td>Strong</td>
          </tr>
          <tr>
            <th scope="row">CSS3</th>
            <td>Comfortable</td>
          </tr>
        </tbody>
      </table>
    </section>
  </main>
</body>
</html>
```

```

        <tr>
            <th scope="row">Git & GitHub</th>
            <td>Comfortable</td>
        </tr>
        <tr>
            <th scope="row">JavaScript</th>
            <td>Beginner</td>
        </tr>
    </tbody>
</table>
</section>

<!-- Contact Form Section -->
<section id="contact" class="card">

    <h2>Contact Me</h2>
    <form action="#" method="post">
        <div>
            <label for="name">Full Name *</label>
            <input type="text" id="name" name="name" required />
        </div>
        <div>
            <label for="email">Email *</label>
            <input type="email" id="email" name="email" required placeholder="name@example.com" />
        </div>
        <div>
            <label for="phone">Phone</label>
            <input type="tel" id="phone" name="phone" />
        </div>

        <fieldset>
            <legend>Preferred Track</legend>
            <label><input type="radio" name="track" value="frontend" checked /> Frontend</label>
            <label><input type="radio" name="track" value="backend" /> Backend</label>
        </fieldset>
        <div>
            <label for="topic">Topic</label>
            <select id="topic" name="topic">
                <option value="">Choose one</option>
                <option value="help">Help</option>
                <option value="feedback">Feedback</option>
            </select>
        </div>
        <div>

            <label for="msg">Message</label>
            <textarea id="msg" name="message" rows="4"></textarea>
        </div>
        <div>
            <label><input type="checkbox" name="terms" required /> I accept the terms</label>
        </div>
        <button type="submit">Send</button>
    </form>
</section>
</main>

<aside>
    <h3>Lab Hours</h3>
    <p>Monday & Wednesday: 2 PM - 4 PM</p>

```

```

</aside>

<!-- Footer -->
<footer>
  <p>&copy; 2026 Muhammad Tayyab (023-24-0118)</p>
  <p>
    <a href="https://github.com" target="_blank" rel="noopener noreferrer">GitHub</a> |
    <a href="https://w3.org" target="_blank" rel="noopener noreferrer">W3C</a>
  </p>
</footer>
</body>
</html>

```

4. Form Testing (Activity D)

- **Label Click Test:** Clicking on the text label 'Full Name *' immediately placed the cursor inside the text box because `<label for="name">` matches `<input id="name">`. When I removed the `for` attribute, clicking the label did nothing.
- **Required Validation:** When I tried to submit the form without filling the required name and email fields, the browser blocked submission with a tooltip saying 'Please fill out this field'.
- **External Link Security:** Added `rel="noopener noreferrer"` to links with `target="_blank"` for security.

5. Homework Task 2: Rewriting Bad HTML (`homework-rewrite.html`)

homework-rewrite.html

```

<!-- Refactored with semantic HTML -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>My Blog</title>
</head>
<body>
  <header>
    <h1>My Blog</h1>
    <nav>
      <a href="/">Home</a>
    </nav>
  </header>
  <main>
    <article class="post">
      <header>
        <h2>First Post</h2>
        <time datetime="2026-08-09">Aug 9 2026</time>
      </header>
      <p>Hello world</p>
    </article>
  </main>
</body>
</html>

```

6. Homework Task 3 & 4 (Second Page & Reflection)

- **Task 3:** Created `contact.html` with a form and linked `profile.html` and `contact.html` to each other in both nav bars and footers.
- **Task 4 (Reflection):**
 1. I was going to use a `<div>` for the about section, but I used `<article>` instead because it represents a standalone piece of content.
 2. For extra sidebar notes, I used `<aside>` instead of a `div`.
 3. Adding `for` and `id` to all labels made the form much easier to click on.
 4. I used `<table>` only for the skills data with `scope="col"` and `scope="row"` as taught.
 5. I tested the page on `validator.w3.org` and got a green pass with 0 errors.

Web Engineering — Lab 03 Report

CSS Fundamentals — Box Model, Selectors & Specificity

Student Name: Muhammad Tayyab	Course: Web Engineering (5th Semester)
CMS ID: 023-24-0118	Lab: Week 03: CSS Fundamentals

1. Lab Summary

In Week 03, we learned how CSS styles HTML elements. We moved inline styles into an external styles.css file, explored the 4 layers of the Box Model (content, padding, border, margin), tested different CSS selectors in the scavenger hunt, and experimented with CSS specificity and cascade order to see which rules win when they conflict.

2. Activity A: Inline to External Stylesheet

We took inline-start.html which had inline style="..." attributes everywhere, removed the inline styles, and moved them into an external styles.css linked in the <head> with <link rel="stylesheet" href="styles.css">.

Activity A: styles.css
<pre>/* styles.css */ body { background: #f4f1ea; font-family: Georgia, serif; } h1 { color: navy; font-size: 2rem; } p { color: #333; font-size: 1.125rem; line-height: 1.6; }</pre>

3. Activity B: The CSS Box Model

Every element in HTML is a box with 4 layers:

1. **Content:** The actual text or image (width, height).
2. **Padding:** Space inside the box between text and border.
3. **Border:** The visible line around the box.
4. **Margin:** Space outside the box between this box and other elements.

What I tested:

- Changed padding to 40px: the text moved further inside away from the border.
- Changed border to 8px solid navy: the outline became thicker.
- Changed margin to 40px: the space between Box 1 and Box 2 grew larger.
- In DevTools (F12), the box model diagram matched these values exactly.

4. Activity C: Selector Scavenger Hunt (7 Rules)

We applied 7 different selectors in styles.css and checked their results on the page:

#	Selector Type	CSS Rule	What happened on page
1	Tag	h1 { color: navy; font-size: 2rem; }	Only the became navy and bigger.
2	Class	.note { color: teal; font-weight: bold; }	Only the class paragraph became teal and bold.
3	ID	#alert { background: #fff3cd; padding: 8px; }	Only the #alert paragraph got a yellow box.
4	Attribute	input[type="email"] { border: 2px solid teal; }	Only the email input got a teal border; text input stayed normal.

5	Hover	a:hover { color: crimson; text-decoration: underline; }	Links turn crimson when mouse points at them.
6	Odd	li:nth-child(odd) { background: #e3f2fd; }	1st and 3rd list items got light blue background.
7	Even	li:nth-child(even) { background: #fff3e0; }	2nd and 4th list items got light orange background.

5. Activity D: Specificity Duel & Cascade

We tested 4 conflicting rules on `<main id="main"><p class="intro">I am the intro paragraph.</p></main>`:

- `p { color: green; }` — Specificity: (0, 0, 1)
- `.intro { color: blue; }` — Specificity: (0, 1, 0)
- `p.intro { color: purple; }` — Specificity: (0, 1, 1)
- `#main p { color: navy; }` — Specificity: (1, 0, 1)

Observations from DevTools:

1. **All 4 rules active:** The text was **navy** because `#main p` has an ID selector which has the highest specificity (1,0,1). In DevTools Styles panel, the other three rules had line-throughs.
2. **Commented out `#main p`:** The text turned **purple** because `p.intro` (0,1,1) has higher specificity than `.intro` (0,1,0).
3. **Commented out `p.intro`:** The text turned **blue** because `.intro` (0,1,0) beats the element tag `p` (0,0,1).
4. **Source order test:** When we had `p { color: green; }` and `p { color: crimson; }` with equal specificity, the text was **crimson** because the last rule in the file wins.

6. Key Takeaways

- Always use external CSS files instead of inline styles.
- Use classes for styling and keep specificity flat.
- Don't use IDs in CSS because they have high specificity and are hard to override.
- Avoid using `!important`.